

Pampered Pets

Executive Summary

Quality, Supply-Chain and Continuity Risk of the International Digitalisation Programme

Prepared for the Board

Date: 9 July 2026

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Figures are quoted at the most-likely revenue base; full low/most-likely/high ranges, sources and formulas are in Appendix A and the companion model

1. Overview and recommendation

This summary sets out, in money and probability, how moving Pampered Pets to an international, digitally enabled operation affects the two things its discerning customers value: consistent product quality and uninterrupted availability. The headline is straightforward: the programme is sound and should proceed. Internationalisation raises the likelihood of a quality-affecting event and creates a supply-availability exposure the single shop never carried, but both are manageable to the brand-promise level if the controls below are funded as core requirements. Figures are quoted against a modelled target-scale revenue of about **£7m** a year (the basis and range are in [Appendix A.1](#)).

Actions are ordered by commercial priority.

#	Action	Why it matters commercially
1	Enforce cold-chain integrity end to end: continuous temperature monitoring and logging, validated packaging, and automatic quarantine of any consignment that breaches limits.	The largest single contributor to quality loss and the most direct threat to the brand the new customers are buying into.
2	Adopt a centralised master product specification with cross-hub batch testing and signature-ingredient control, so product is indistinguishable between regions.	Protects brand consistency, the real risk once production is regional rather than single site.
3	Make dual-sourcing and the multi-hub production network the primary supply-availability control and hold a six-week (42-day) global buffer to bridge switchover.	Buffer stock alone cannot cover the month-plus disruptions that occur roughly every 3.7 years; structural redundancy can.
4	Run the online shop active-active across two cloud regions with automated failover and synchronous replication, backed by immutable point-in-time backups.	The only architecture that delivers 24/7/365 trading with sub-one-minute recovery and near-zero data loss.
5	Put regional data-protection and cross-border transfer arrangements in place before any customer data moves between hubs.	A legal precondition of international trading; exposure is amplified by high-profile customers.
6	Select the back-office platform against explicit exit terms (open data formats, portable database, contractual data return). and secure a source-code / SaaS escrow as a break-glass control.	Avoids a costly future dependency in the system that becomes the operational core.

Table 1 List of action to mitigate risk of impact on quality and availability

2. Approach to the estimates

The figures are produced with a recognised quantitative risk method rather than a colour-coded matrix, because the Board is committing capital and needs exposure in money and probability.

Each risk is broken into how often the triggering event occurs, how likely it is to cause real harm, and what it would cost, following the Factor Analysis of Information Risk taxonomy (Open Group, 2021), with loss magnitudes expressed as a share of revenue so they move with the business as it scales.

Those inputs are run through a Monte Carlo simulation (Metropolis, 1987; Eckstein and Riedmueller, 2002), producing a full range of outcomes rather than a single average that would hide the costly but unlikely cases (Cox, 2008; Hubbard, 2020).

Because the international operation has no trading history, the inputs are calibrated expert estimates, stated openly in [Appendix A](#) and tightening as the business builds its own data (Aven and Thekdi, 2025). Where knowledge is weakest, this is said plainly, so confidence is placed only where it is earned.

3. Risk to product quality

Five events dominate the risk to delivered quality once production and distribution span three regions:

- **Cold-chain excursion**
- **Transit delay and spoilage**
- **Contamination or adulteration**
- **Supplier quality drift**
- **Inter-hub quality variance**

The probability that at least one quality-affecting deviation occurs somewhere in the network in a year is high, about **94%**. This is expected for any global chain and is not in itself alarming; the management question is severity and consistency, not occurrence. The probability of a serious, recall-grade event is about **17%** a year, and of a brand-material inter-hub inconsistency reaching customers about **34%**.

The simulated cost of quality loss is about **£175,000** in a typical year and about **£460,000** in a plausible bad year (P90: the 90th percentile); the gap between the two is the recall-driven tail that an average alone would conceal ([Appendix A.2](#)).

These figures precede the recommended controls. Cold-chain monitoring and validated packaging cut the largest contributor; a centralised master specification with cross-hub testing addresses inter-hub variance directly, since that risk is one of

process design rather than control failure and is not removed by supplier auditing alone. The residual position after controls should be re-run and presented as a before-and-after comparison.

4. Risk to supply availability

These physical disruptions are inherent to any international supply chain; digitalisation does not cause them but changes the risk profile. The automated warehouses depend on the ERP supply-chain module to orchestrate flow and reroute around them, so that module now sits on the critical path: its failure would halt fulfilment even when the physical chain is intact.

Supply interruption is modelled as the chance a disruption outlasts the stock in reserve. The decisive finding is that buffer stock alone cannot solve this, because significant disruptions lasting a month or longer occur roughly every 3.7 years on average (McKinsey Global Institute, 2020).

At a 21-day buffer the annual probability of at least one stockout is about one-in-two; a six-week (42-day) buffer brings it to roughly one-in-nine, and only a very large buffer pushes it lower ([Appendix A.3](#)).

Inventory therefore mitigates but cannot eliminate the exposure. The primary controls are structural: dual sourcing every critical ingredient and the multi-hub production network ([Section 5.1](#)), so a disruption at one source or hub is absorbed by another, with the buffer bridging the switchover.

The principal disruption sources and their indicative annual probabilities are below.

Supply disruption	P (per year)	Primary mitigation
Last-mile delivery failure	0.96	Multiple carriers; click-and-collect fallback
Port or freight congestion	0.79	Dual-region freight; buffer bridge
Fuel or energy price shock	0.68	Indexed contracts; regional sourcing
Logistics-provider data or insider incident	0.56	Vetting; contractual security terms; least privilege
Natural disaster at a warehouse hub	0.49	Geographic node diversity; multi-hub cover
Geopolitical event or maritime blockade	0.38	Dual-region sourcing; scenario planning

Table 2 List of main supply chain disruptions and suggested mitigation

Important note: Pet food is subject to country-specific feed and import controls. Confirming compliance in each destination market is a precondition of entry, not a later step, since a blocked or recalled shipment is both a quality and an availability failure.

5. Continuity and disaster recovery

Resilience is required in three distinct places:

- the **physical production hubs** that make the product,
- the **core systems** (ERP/WMS) that orchestrate them, and
- the **digital platform** that sells it.

They have different failure modes and different recovery designs and are treated separately.

5.1 Production continuity

Assumed network design: three regional production hubs (Americas, EMEA, APAC), plus a wider set of automated distribution warehouses. Each hub carries about 150 percent of its own steady-state demand, so if one hub is lost the two survivors each take a share of its output and continue to meet global demand. Single-hub loss is therefore an operational inconvenience, not an availability event, by design.

Product availability is threatened only when two or more hubs are disrupted at once. With an indicative annual per-hub disruption probability of **5%**, the probability of two or more hubs being down simultaneously is about **0.7%** a year ([Appendix A.4](#)).

The caveat is that this assumes hubs fail independently. Under correlated global shocks, a pandemic or a region-spanning geopolitical event, they do not, and that correlated tail, not routine single-hub failure, is the genuine residual risk. It is addressed by geographic and supplier diversity across hubs and the [Section 4](#) buffer.

5.2 Core systems (ERP/WMS)

The hub redundancy in [5.1](#) protects against physical loss but assumes the system running the hubs is available. In automated warehouses a hub with power, staff and stock but no working ERP/WMS cannot produce, so the control system is a single dependency sitting beneath all three hubs at once: its failure is a correlated, whole-network production outage that hub headroom cannot absorb. The ERP/WMS

therefore needs the same active-active, automated-failover and immutable-backup design as the online shop ([5.3](#)), but for a more fundamental reason: the online platform protects revenue and reputation, whereas this protects the ability to make product at all.

5.3 Online shop recovery

The online shop must trade continuously, recover within one minute, and lose no more than one minute of data. These objectives rule out conventional cold or warm standby sites, whose recovery runs to hours or days, and make even a manually triggered hot standby marginal, since detection and switchover alone can consume the entire one-minute budget.

The defensible design is active-active across two cloud regions: both serve live traffic behind a global load balancer with health checks, so a region failure removes capacity rather than causing an outage and recovery is effectively continuous. On standard availability mathematics this yields well under a minute of unplanned downtime a year, inside the 99.99 to 99.999 percent band the requirement implies (Andrade et al., 2017; [Appendix A.5](#)).

The direct cost of downtime is small: at the modelled revenue an hour offline forgoes only about **£320** of online sales, and a year at 99.99 percent availability about **£280**. The justification is reputational, not transactional: a visible outage to a King or Head of State is a brand event, drawing on the same exposure quantified in [Section 7](#), and the investor has set continuous availability as a requirement. The recommendation is to meet, not relax, the objective, sizing the platform to actual traffic to reach the cheapest configuration that meets it (Andrade et al., 2017).

The sub-one-minute data-loss target requires synchronous replication between regions, giving near-zero loss. Replication is not a backup, because it copies corruption and ransomware as faithfully as legitimate data; the design therefore adds immutable, point-in-time backups (three copies, two media, one offsite, one offline, zero recovery errors after testing). Recovery is credible only once exercised, so a full regional failover should be rehearsed before launch and annually thereafter.

6. Platform selection and vendor lock-in

Lock-in risk concentrates in the back office, not the storefront. The customer-facing marketplace is comparatively portable; the enterprise resource planning (ERP) system becomes the single source of truth, so leaving it later is costly and disruptive (Kumar, 2024; Badger et al., 2012). That dependency is now deeper still, because the ERP orchestrates the automated warehouses rather than only recording transactions, so exiting it means re-integrating and re-testing physical fulfilment, raising both switching cost and operational exposure. Sized as an expected exit cost (probability of forced migration times migration cost), choosing a low-lock-in platform over a deeply proprietary one is worth, in expected terms, on the order of **£120,000** ([Appendix A.6](#)).

This is an order-of-magnitude steer, not a precise figure, but shows the difference is material enough to make exit terms a selection criterion. Selection should weigh total cost of exit, not only entry, and avoid provider-specific services that cannot be reproduced elsewhere. A source-code or, for cloud, SaaS escrow is a complementary break-glass control, releasing the code and, for cloud, data and runtime if the vendor fails or abandons the product; its value depends on the system using common, maintainable technology, since proprietary or legacy code cannot be staffed, so escrow reinforces the preference for open platforms rather than replacing it.

7. Data protection and regulatory requirements

Holding customer data across three regions brings several regimes into play at once, and they differ in ways that constrain where data may be processed. These are launch-gating business requirements:

- **United Kingdom (UK GDPR and Data Protection Act 2018)**
- **European Economic Area (EU GDPR)**
- **United States (California CCPA/CPRA)**
- **China (PIPL)**

A Data Protection Impact Assessment should precede launch, and the recovery and incident plans must support breach notification within the tightest applicable window (72 hours from awareness under the UK and EU regimes). The exposure sizes like the operational risks: an indicative annual probability of a reportable breach of **15%** and an expected cost per breach of around **£250,000 to £400,000**, dominated not by the regulatory fine, modest at this turnover, but by the reputational damage of a disclosure involving high-profile customers ([Appendix A.7](#)).

The practical conclusion is that restricting and logging access to prominent customers' data is a priority control.

8. Conclusion

The programme is viable and the recovery objectives achievable. The international supply chain converts a previously low-quality risk into a material, recurring one, but the modelling shows the exposure is concentrated, long tailed and controllable. Cold-chain integrity, a centralised specification with cross-hub testing, dual-sourcing and multi-hub redundancy backed by a six-week buffer, an active-active online platform and regionally appropriate data-protection controls reduce the residual risk to a level consistent with the world-class quality the new customers expect. The estimates are calibrated judgements, anchored to data where it exists and to reasoned judgement where it does not, to be refined as the business builds its own record.

Word count: 2,035 (main content excluding titles, table of content, references and appendix.
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References

- Andrade, E., Nogueira, B., Matos, R., Callou, G. and Maciel, P. (2017)** 'Availability modeling and analysis of a disaster-recovery-as-a-service solution', *Computing*, 99(9), pp. 929-954. doi:10.1007/s00607-017-0539-8.
- Aven, T. and Thekdi, S. (2025)** *Risk Science: An Introduction*. 3rd edn. Oxon: Routledge.
- Badger, M., Grance, T., Patt-Corner, R. and Voas, J. (2012)** *Cloud Computing Synopsis and Recommendations*. Special Publication 800-146. Gaithersburg, MD: National Institute of Standards and Technology. doi:10.6028/NIST.SP.800-146.
- Cox, L.A. (2008)** 'What's wrong with risk matrices?', *Risk Analysis*, 28(2), pp. 497-512.
- Eckstein, J. and Riedmueller, S.T. (2002)** 'YASAI: yet another add-in for teaching elementary Monte Carlo simulation in Excel', *INFORMS Transactions on Education*, 2(2), pp. 12-26. doi:10.1287/ited.2.2.12.
- Food Standards Agency (2025)** *Food and allergy alerts*. London: Food Standards Agency. Available at: <https://www.food.gov.uk/news-alerts> (Accessed: 29 June 2026).
- Hubbard, D.W. (2020)** *The Failure of Risk Management: Why It's Broken and How to Fix It*. 2nd edn. Hoboken, NJ: Wiley.
- IBM Security (2025)** *Cost of a Data Breach Report 2025*. Armonk, NY: IBM Corporation. Available at: <https://www.ibm.com/reports/data-breach> (Accessed: 29 June 2026).
- IBISWorld (2025)** *Garden Centres and Pet Shops in the UK*. London: IBISWorld. Available at: <https://www.ibisworld.com/> (Accessed: 28 June 2026).
- ICO (2024)** *Guide to the UK General Data Protection Regulation*. Wilmslow: Information Commissioner's Office. Available at: <https://ico.org.uk/> (Accessed: 26 May 2026).
- Kumar, A. (2024)** *Cloud Vendor Lock-In: Identify, Strategies and Mitigate*. Seminar paper. Würzburg: Julius-Maximilians-Universität Würzburg.
- McKinsey Global Institute (2020)** *Risk, resilience, and rebalancing in global value chains*. New York: McKinsey & Company. Available at: <https://www.mckinsey.com/> (Accessed: 29 June 2026).
- Metropolis, N. (1987)** 'The beginning of the Monte Carlo method', *Los Alamos Science*, 15, pp. 125-130.
- Office for National Statistics (2025)** *Annual turnover of stores specialising in flowers, plants, pet animals and pet food, UK, 2008 to 2023*. Accessed via Statista. Available at: <https://www.statista.com/statistics/429527/> (Accessed: 30 June 2026).
- Open Group (2021)** *Risk Analysis (O-RA), Version 2.0.1. Reading: The Open Group*.
- Pet Gazette (2025)** *UK Petcare Retail Market Report 2025: Top 30 Petcare Retailers*. Available at: <https://www.petgazette.biz/> (Accessed: 30 June 2026).

Appendix A: Assumptions and calculations

Inputs are calibrated estimates expressed as PERT (minimum, most likely, maximum) distributions, sampled by Monte Carlo simulation. The mean of a PERT distribution is $(\text{minimum} + 4 \times \text{most likely} + \text{maximum}) / 6$. The companion workbook, ***U11 - Individual Project - Executive Summary (Risk Model).xlsx*** – holds every input as an editable cell and recalculates all figures live.



A.1 Revenue Base

The revenue base is a modelled assumption for the funded, target-scale operation, constructed from published anchors. It is not a figure taken directly from any single source and is stated as a range.

Anchors: the UK garden-and-pet specialised-store sector turned over about £6.2bn in 2023 (Office for National Statistics, 2025) and is forecast near £6.8bn for 2025-26 (IBISWorld, 2025); a premium multi-store comparator implies roughly £600k of turnover per store in a high-service format (Pet Gazette, 2025). Scaled to a funded three-hub, multi-channel operation:

Base assumption	Low	Most likely	High
Annual revenue	£3m	£7m	£15m
Gross margin	35%	45%	55%
Online share of revenue	25%	40%	60%

Derived units at the most-likely base:

- revenue per day **£19,178**;
- gross margin per day **£8,630**;
- online revenue per minute **£5.33** (per hour £320).

Gross margin per day is the unit used for lost-sales risks (stockouts, downtime), because the economic loss of unsold product is the margin forgone, not full revenue. Recalls are modelled as a share of revenue plus disposal and regulatory costs, because they destroy more than margin.

A.2 Product quality

For each risk R1 to R5: rate = $F \times V$, where F is annual event frequency and V is the probability quality is harmed given the event. The combined rate is the sum of the five rates. The probability of at least one quality-affecting event in a year is 1 minus “e” raised to the negative combined rate (the probability that none occur, subtracted from one).

Expected annual loss is the sum across risks of rate $\times$ magnitude, where magnitude is a share of revenue. The 90th-percentile loss is read from the simulated distribution and has no closed form, which is the reason for simulating.

Risk	F	V	rate	mag mean (% / £)	Loss = rate x mag (£)
R1 Cold-chain excursion	3.50	0.32	1.11	0.38% / £27k	£30k
R2 Transit delay / spoilage	2.25	0.27	0.60	0.53% / £37k	£22k
R3 Contamination / recall	0.32	0.60	0.19	5.3% / £373k	£71k
R4 Supplier quality drift	1.22	0.42	0.51	0.45% / £31k	£16k
R5 Inter-hub variance	0.68	0.61	0.42	1.25% / £87k	£36k
Combined			2.82		£175k

Worked results (at £7m revenue). $P(\geq 1 \text{ quality event}) = 1 - e^{(-2.82)} = 0.94$. Serious/recall event (R3 alone) = $1 - e^{(-0.19)} = 0.17$. Inter-hub variance (R5) = $1 - e^{(-0.42)} = 0.34$. Expected annual loss = £175k; simulated P90 $\approx$ £460k (the live model reproduces this and varies slightly each run because the draws are random). Expected loss scales with revenue: about £75k at £3m, £376k at £15m.

Evidence basis: R3 (contamination / recall) frequency is anchored to recall records, which show pet-food recalls are rare for any single producer (Food Standards Agency, 2025; corroborated by US FDA pet-food recall data), scaled down from a generic placeholder but kept above a single-UK-producer baseline to reflect the new international supplier base.

R1, R2, R4 and R5 are calibrated expert estimates with weak-to-moderate supporting knowledge, as no firm-specific public data exists; they are to be re-calibrated against operating data (Aven and Thekdi, 2025; Hubbard, 2020).

A.3 Supply availability

Significant disruptions arrive at frequency PERT(0.8, 1.5, 3.5) per year; duration = PERT(3, 10, 80) days. Buffer-exceeding disruptions form a thinned Poisson process at rate (frequency $\times$ P(duration > buffer)), so $P(\geq 1 \text{ stockout/yr}) = 1 - e^{-(\text{that rate})}$. These inputs are anchored to McKinsey Global Institute (2020): the model produces about 0.36 month-plus (over 30-day) disruptions per year, consistent with the reported one every 3.7 years (0.27 per year), the small excess reflecting a new, multi-party international chain. Results: 14-day buffer 0.66; 21-day 0.51; 28-day 0.35; 42-day 0.11; 60-day below 0.01. Each day of stockout beyond the buffer costs about £8,630 in lost gross margin. The key implication is that buffer stock cannot eliminate the exposure on its own; dual-sourcing and the multi-hub network are the primary controls, with the buffer bridging switchover.

Strength of knowledge: moderate (data-anchored frequency, judgement-based mapping to this business).

A.4 Production hub availability

Let q be the annual probability one hub suffers a capacity-removing disruption. With three hubs each holding enough headroom that any two cover all three, availability is breached only if two or more are down at once: $P(\geq 2 \text{ of } 3) = 3q^2(1 - q) + q^3$. At $q = 0.05$ this is 0.7%; at $q = 0.03$, 0.3%; at $q = 0.08$, 1.8%. Assumes independent hub failures; correlated global shocks break this and form the dominant tail risk.

Strength of knowledge: weak.

A.5 Online-shop availability

Availability $A = \text{MTBF} / (\text{MTBF} + \text{MTTR})$, where MTBF (Mean Time Between Failures) is the average time a service runs before failing, and MTTR (Mean Time To Recover) is the average time to restore it.

Two independent regions in active-active give combined availability $A_t = 1 - (1 - A)^2$. At a per-region A of 0.9995 (about 263 minutes downtime per year), $A_t \approx 0.9999998$, under one minute per year. Correlated dependencies (DNS, control plane) reduce this, so a 99.99 to 99.999 percent target (5 to 53 minutes per year) is the realistic commitment. Sub-one-minute recovery is met because both regions are already live; sub-one-minute data loss is met by synchronous replication. Direct lost online sales at

99.99 percent (about 53 minutes/year) $\approx$ £280; the requirement is justified by reputation, not direct revenue.

A.6 Vendor lock-in

Expected exit-cost difference = $P(\text{forced migration within horizon}) \times (\text{switch cost on a high-lock-in platform} - \text{switch cost on a low-lock-in platform})$.

Illustrative: $P = 0.30$; high-lock-in switch cost $\text{PERT}(350, 600, 1000) = \text{£}625\text{k}$ mean; low-lock-in $\text{PERT}(100, 200, 400) = \text{£}217\text{k}$ mean. Difference = $0.30 \times (625 - 217) = \text{£}122\text{k}$. The higher ranges reflect the ERP/WMS orchestrating the automated warehouses, so exit requires re-integrating physical fulfilment. A code or SaaS escrow (code plus, for cloud, data and runtime) is a complementary break-glass control whose value depends on the system using common, maintainable technology.

Strength of knowledge: weak; order-of-magnitude decision aid only.

A.7 Data-breach exposure

Annualised loss = $P(\text{reportable breach/yr}) \times \text{expected loss per breach}$. Illustrative: $P = 0.15$; loss per breach = $\text{PERT}(2\%, 4\%, 8\%)$ of revenue $\approx \text{£}303\text{k}$ mean ($P90 \approx \text{£}411\text{k}$), comprising notification, remediation, regulatory penalty and reputational damage, the last dominant given high-profile customers. Annualised mean $\approx \text{£}46\text{k}$. Evidence basis: anchored to IBM Security (2025), which reports a UK average breach cost of £3.29m; that average reflects large enterprises with far bigger record counts, so the SME estimate here sits well below it, with the reputational tail the main uncertainty.